

Contact

Office: 267 Bloomberg Hall, Institute for Advanced Study, Princeton, NJ 08544

Employment

2022 - 2027

2022 - 2025

Education

2017 - 2022

M.S. in Physics

Ph.D. in Physics

Advisor: Shinsei Ryu

Princeton University

2021 - 2022

Visiting Student Research Collaborator

Trailing Student

Massachusetts Institute of Technology

2020 - 2021

Visiting Student

Host: Hong Liu

Colgate University

2013 - 2017

A.B. in Astronomy/Physics *magna cum laude*

King's College London

2016

Study Abroad Student

Publications

1. M. S. Klinger, **J. Kudler-Flam**, G. Satishchandran, “Generalized Entropy is von Neumann Entropy II: The complete symmetry group and edge modes” arXiv:2601.07910 [hep-th]. Under Review.
2. Akash Vijay, Ayush Raj, **J. Kudler-Flam**, Benot Vermersch, Andreas Elben, Laimei Nie, “Analytically Continuing the Randomized Measurement Toolbox” arXiv:2511.02912 [quant-ph]. Under Review.
3. **J. Kudler-Flam**, E. Witten, “Emergent Mixed States for Baby Universes and Black Holes” arXiv:2510.06376 [hep-th]. Under Review.
4. A. Herderschnee, **J. Kudler-Flam**, “Stringy algebras, stretched horizons, and quantum-connected wormholes ” arXiv:2510.01556 [hep-th]. Under Review.
5. A. Blommaert, **J. Kudler-Flam**, E. Y. Urbach, “Absolute entropy and the observer’s no-boundary state” arXiv:2505.14771 [hep-th]. Accepted To Journal of High Energy Physics.
6. **J. Kudler-Flam**, V. Narovlansky, N. Sopenko, “Optimal bound on long-range distillable entanglement ” arXiv:2504.02926 [quant-ph]. Phys. Rev. Lett. 135, 060403.
7. **J. Kudler-Flam**, G. Penington, “It costs nothing to teleport information into a black hole” arXiv:2504.01058 [hep-th]. International Journal of Modern Physics D. Awarded second prize in 2025 Gravity Research Foundation Essay Contest.
8. **J. Kudler-Flam**, K. Prabhu, G. Satishchandran, “Vacua and infrared radiation in de Sitter quantum field theory ” arXiv:2503.19957 [hep-th]. Under Review.

9. D.L. Danielson, **J. Kudler-Flam**, G. Satishchandran, R.M. Wald, “How to Minimize the Decoherence Caused by Black Holes” arXiv:2501.04773 [hep-th]. Phys. Rev. D 112, 025012
10. H.J. Chen and **J. Kudler-Flam**, “Free Independence and the Noncrossing Partition Lattice in Dual-Unitary Quantum Circuits” arXiv:2409.17226 [hep-th]. Phys. Rev. B 111, 014311
11. X. Dong, **J. Kudler-Flam**, and P. Rath, “Entanglement Negativity and Replica Symmetry Breaking in General Holographic States” arXiv:2409.13009 [hep-th]. Journal of High Energy Physics 01 022 (2025)
12. **J. Kudler-Flam**, S. Leutheusser, and G. Satishchandran, “Algebraic Observational Cosmology” arXiv:2406.01669 [hep-th]. Under Review.
13. **J. Kudler-Flam**, S. Leutheusser, A. A. Rahman, G. Satishchandran, and A. Speranza, “A covariant regulator for entanglement entropy: proofs of the Bekenstein bound and QNEC” arXiv:2312.07646 [hep-th]. Phys. Rev. D 111, 105001
14. X. Dong, **J. Kudler-Flam**, and P. Rath, “A Modified Cosmic Brane Proposal for Holographic Renyi Entropy” arXiv:2312.04625 [hep-th]. Journal of High Energy Physics 06 120 (2024).
15. **J. Kudler-Flam**, S. Leutheusser, and G. Satishchandran, “Generalized Black Hole Entropy is von Neumann Entropy” arXiv:2309.15897 [hep-th]. Phys. Rev. D 111, 025013
16. **J. Kudler-Flam**, M. Nozaki, T. Numasawa, S. Ryu, and M.T. Tan, “Bridging two quantum quench problems – local joining quantum quench and Möbius quench – and their holographic dual descriptions,” arXiv:2309.04665 [hep-th]. Journal of High Energy Physics 08 213 (2024).
17. **J. Kudler-Flam**, L. Nie, and A. Vijay, “Rényi mutual information in quantum field theory, tensor networks, and gravity” arXiv:2308.08600 [hep-th]. Journal of High Energy Physics 06 195 (2024).
18. G. Cipolloni and **J. Kudler-Flam**, “Non-Hermitian Hamiltonians Violate the Eigenstate Thermalization Hypothesis,” arXiv:2303.03448 [cond-mat.stat-mech]. Phys. Rev. B 109, L020201 *Editor’s Suggestion*.
19. Y. Liu, **J. Kudler-Flam**, and K. Kawabata, “Symmetry Classification of Typical Quantum Entanglement,” arXiv:2301.07778 [cond-mat.mes-hall]. Phys. Rev. B 108, 085109.
20. Y. Liu, Y. Kusuki, R. Sohal, **J. Kudler-Flam**, and S. Ryu, “Multipartite entanglement in two-dimensional chiral topological liquids,” arXiv:2301.07130 [cond-mat.str-el]. Phys. Rev. B 109, 085108.
21. **J. Kudler-Flam** and Y. Kusuki, “On Quantum Information Before the Page Time,” arXiv:2212.06839 [hep-th]. Journal of High Energy Physics 08 189 (2022).
22. **J. Kudler-Flam** “Rényi Mutual Information in Quantum Field Theory,” arXiv:2211.01392 [hep-th]. Phys. Rev. Lett. 130, 021603.
23. G. Cipolloni and **J. Kudler-Flam** “Entanglement Entropy of Non-Hermitian Eigenstates and the Ginibre Ensemble,” arXiv:2206.12438 [cond-mat.stat-mech]. Phys. Rev. Lett. 130, 010401.
24. **J. Kudler-Flam** and P. Rath, “Large and Small Corrections to the JLMS Formula from Replica Wormholes,” arXiv:2203.11954 [hep-th]. Journal of High Energy Physics 08 189 (2022).
25. S. Vardhan, **J. Kudler-Flam**, H. Shapourian, and H. Liu, “Mixed-state entanglement and information recovery in thermalized states and evaporating black holes,” arXiv:2112.00020 [hep-th]. Journal of High Energy Physics 01 064 (2023).
26. Y. Liu, R. Sohal, **J. Kudler-Flam**, and S. Ryu, “Multipartitioning topological phases by vertex states and quantum entanglement,” arXiv:2110.11980 [cond-mat.str-el]. Phys. Rev. B 105, 115107.

27. S. Vardhan, **J. Kudler-Flam**, H. Shapourian, and H. Liu, “Bound entanglement in thermalized states and black hole radiation,” arXiv:2110.02959 [hep-th]. Phys. Rev. Lett. 129, 061602.
28. **J. Kudler-Flam**, V. Narovlansky, and S. Ryu, “Negativity Spectra in Random Tensor Networks and Holography,” arXiv:2109.02649 [hep-th]. Journal of High Energy Physics 02 076 (2022).
29. **J. Kudler-Flam**, V. Narovlansky, and S. Ryu, “Distinguishing Random and Black Hole Microstates,” arXiv:2108.00011 [hep-th]. Phys. Rev. X Quantum 2, 040340.
30. **J. Kudler-Flam**, R. Sohal, and L. Nie, “Information Scrambling with Conservation Laws,” arXiv:2107.04043. SciPost Phys. 12, 117 (2022).
31. **J. Kudler-Flam**, “Relative Entropy of Random States and Black Holes,” arXiv:2102.05053 [hep-th]. Phys. Rev. Lett. 126, 171603.
32. H. Shapourian, S. Liu, **J. Kudler-Flam**, and A. Vishwanath, “Entanglement negativity spectrum of random mixed states: A diagrammatic approach,” arXiv:2011.01277 [cond-mat.str-el]. Phys. Rev. X Quantum 2, 030347.
33. **J. Kudler-Flam**, Y. Kusuki, and S. Ryu, “The quasi-particle picture and its breakdown after local quenches: mutual information, negativity, and reflected entropy,” arXiv:2008.11266 [hep-th]. Journal of High Energy Physics 03 146 (2021).
34. **J. Kudler-Flam**, M. Nozaki, S. Ryu, and M. Tian Tan, “Entanglement of Local Operators and the Butterfly Effect,” arXiv:2005.14243 [hep-th]. Phys. Rev. Res. 3, 033182.
35. M. Asrat and **J. Kudler-Flam**, “ $TT\bar{T}$, the entanglement wedge cross section, and the breakdown of the split property,” arXiv:2005.08972 [hep-th]. Phys. Rev. D 102, 045009.
36. I. MacCormack, M. Tian Tan, **J. Kudler-Flam**, and S. Ryu, “Operator and entanglement growth in non-thermalizing systems: many-body localization and the random singlet phase,” arXiv:2001.08222 [cond-mat.str-el]. Phys. Rev. D 104, 214202.
37. **J. Kudler-Flam**, Y. Kusuki, and S. Ryu, “Correlation measures and the entanglement wedge cross-section after quantum quenches in two-dimensional conformal field theory,” arXiv:2001.05501 [hep-th]. Journal of High Energy Physics 04, 74 (2020).
38. **J. Kudler-Flam**, L. Nie, and S. Ryu, “Conformal field theory and the web of quantum chaos diagnostics,” arXiv:1910.14575 [hep-th]. Journal of High Energy Physics 01, 175 (2020).
39. **J. Kudler-Flam**, H. Shapourian, and S. Ryu, “The negativity contour: a quasi-local measure of entanglement for mixed states,” arXiv:1908.07540 [hep-th]. SciPost Phys. 8, 063 (2020).
40. Y. Kusuki, **J. Kudler-Flam**, and S. Ryu, “Derivation of holographic negativity in $\text{AdS}_3/\text{CFT}_2$,” arXiv:1907.07824 [hep-th]. Phys. Rev. Lett. 123, 131603.
41. **J. Kudler-Flam**, M. Nozaki, S. Ryu, and M. Tian Tan, “Quantum vs. classical information: operator negativity as a probe of scrambling,” arXiv:1906.07639 [hep-th]. Journal of High Energy Physics 01, 031 (2020).
42. **J. Kudler-Flam**, I. MacCormack, and S. Ryu, “Holographic entanglement contour, bit threads, and the entanglement tsunami,” arXiv:1902.04654 [hep-th]. J. Phys. A: Math. Theor. 52 (2019) 325401.
43. **J. Kudler-Flam** and S. Ryu, “Entanglement negativity and minimal entanglement wedge cross-sections in holographic theories,” arXiv:1808.00446 [hep-th]. Phys. Rev. D 99, 106014.
44. J. Chen and **J. Kudler-Flam**, “Laplacian growth & sandpiles on the Sierpinski gasket: limit shape universality and exact solutions,” arXiv:1807.08748 [math-ph]. Annales de l’Institut Henri Poincaré D 7 (4).

Invited Talks

1. Stanford University, Stanford, CA
SITP Monday Colloquium
Large fluctuations in the large-N limit
Mar. 2026
2. OIST, Okinawa, Japan
Qubits and Spacetime Unit Group Seminar
Generalized Entropy is von Neumann Entropy
Mar. 2026
3. University of Michigan, Ann Arbor, MI
LITP Workshop on Quantum Black Holes
Large fluctuations in the large-N limit
Mar. 2026
4. Institute for Advanced Study, Princeton, NJ
Workshop on Quantum Aspects of Black Holes and Spacetime
Emergent Mixed States for Baby Universes and Black Holes
Dec. 2025
5. Steklov Mathematical Institute, Moscow, Russia
Frontiers in Holography Workshop
Emergent Mixed States for Baby Universes and Black Holes
Nov. 2025
6. Massachusetts Institute of Technology, Cambridge, MA
String/Gravity Theory Seminar
Stringy algebras, stretched horizons, and quantum-connected wormholes
Nov. 2025
7. Brandeis University, Waltham, MA
Quantum/Gravity Seminar
Emergent Mixed States for Baby Universes and Black Holes
Oct. 2025
8. University of Massachusetts, Amherst, MA
Amherst Center for Fundamental Interactions Seminar
Emergent Mixed States for Baby Universes and Black Holes
Oct. 2025
9. Institute for Advanced Study, Princeton, NJ
Quantum Aspects of Black Holes Meeting
Emergent Mixed States for Baby Universes and Black Holes
Oct. 2025
10. Institute for Advanced Study, Princeton, NJ
Quantum Aspects of Black Holes Meeting
Discussion of the Lorentzian Gravitational Path Integral
Oct. 2025
11. Castello dei Principi Capano, Pollica, Italy
Workshop on Universality in non-equilibrium quantum matter
An Optimal Bound on Long-Range Distillable Entanglement
Sep. 2025
12. Perimeter Institute, Waterloo, Canada
QIQG 2025
Absolute Entropy and the Observer's No-Boundary State
Jun. 2025
13. Institute for Advanced Study, Princeton, NJ
Quantum Aspects of Black Holes Meeting
Absolute Entropy and the Observer's No-Boundary State
Jun. 2025
14. Princeton University, Princeton, NJ
PCTS Year End Celebration
Comments on the identity matrix
Mar. 2025
15. Texas Tech, Lubbock, TX
Quantum Information and Computing Seminar
An optimal bound on long-range distillable entanglement
Apr. 2025
16. Brown University, Providence, RI
Brown Theoretical Physics Center IDEA Seminar Series
Entropy and the Hartle-Hawking State
Apr. 2025

17. Cornell University, Ithaca, NY Feb. 2025
LEPP Theory Seminar
Aspects of the Hartle-Hawking State
18. Institute for Advanced Study, Princeton, NJ Feb. 2025
Physics Group Meeting
An Energy Efficient way to send Qubits into Black Holes
19. University of Massachusetts, Amherst, MA Jan. 2025
NE Gravity Workshop
Black Hole Entropy is von Neumann Entropy
20. Institute for Advanced Study, Princeton, NJ Nov. 2024
Quantum Aspects of Black Holes Meeting
Vacua and Infrared Tails in de Sitter
21. Trinity College, Dublin, Ireland June. 2024
HMI Workshop: CFT and Holography for Heavy and Thermal States and Black Holes
Black Hole Entropy is von Neumann Entropy
22. University of Toronto, Toronto, Canada May. 2024
Theoretical High-Energy Physics Seminar
Algebraic Observational Cosmology
23. University of Chicago, Chicago, IL Mar. 2024
Quantum Aspects of Cosmology Workshop
Algebraic Observational Cosmology
24. Institute for Advanced Study, Princeton, NJ Mar. 2024
Quantum Aspects of Black Holes Meeting
Algebraic Observational Cosmology
25. Caltech University, Pasadena, CA Mar. 2024
High Energy Theory Seminar
Algebras and entropies for quantum field theory, black holes, and inflationary cosmology
26. Institute for Advanced Study, Princeton, NJ Oct. 2023
Physics Group Meeting
Entropy differences in quantum field theory and a rigorous proof of the Bekenstein bound
27. KTH Royal Institute of Technology, Stockholm, Sweden May 2023
Complex Systems and Biological Physics Seminars
Horizons as eavesdroppers: decoherence from soft radiation
28. Institute for Basic Science, Daejeon, Republic of Korea May 2023
Center for Theoretical Physics of Complex Systems Seminar
Non-Hermitian Hamiltonians Violate the Eigenstate Thermalization Hypothesis
29. Purdue University, West Lafayette, IN Apr. 2023
Department of Physics & Astronomy Seminar
Horizons as eavesdroppers: decoherence from soft radiation
30. Oxford University, Oxford, UK Apr. 2023
Sondhi Group Meeting
Horizons as eavesdroppers: decoherence from soft radiation
31. University of California Berkeley, Berkeley, CA Apr. 2023
Berkeley Center for Theoretical Physics String Seminar
Horizons as eavesdroppers: decoherence from soft radiation
32. Stanford University, Stanford, CA Apr. 2023
Stanford Institute for Theoretical Physics Seminar
Horizons as eavesdroppers: decoherence from soft radiation

33. City College of the City University of New York, New York, NY
Theoretical Physics Seminar
Horizons as eavesdroppers: decoherence from soft radiation
Mar. 2023
34. Institute for Advanced Study, Princeton, NJ
Quantum Aspects of Black Holes Meeting
The Information-Disturbance Tradeoff and Horizons as Eavesdroppers
Mar. 2023
35. University of Pennsylvania, Philadelphia, PA
High Energy Theory Seminar
Rényi mutual information in quantum field theory and gravity
Feb. 2023
36. Southeast University, Nanjing, China
2022 Forum for Young Scholars in Mathematics and Mathematical Physics
Rényi mutual information in quantum field theory
Dec. 2022
37. Institute for Advanced Study, Princeton, NJ
High Energy Theory Seminar
Quantum Information before the Page Time
Sep. 2022
38. Princeton University, Princeton, NJ
Princeton Center for Theoretical Science Annual Retreat
Distinguishability and its use: from spin chains to black holes
Sep. 2022
39. University of Amsterdam, Amsterdam, Netherlands
Institute of Physics String Theory Seminar
On the Information Content of Black Hole Radiation
Mar. 2022
40. Steklov Mathematical Institute, Moscow, Russia
Frontiers in Holography Workshop
Distinguishability in Random States, Eigenstates, and Gravity
Dec. 2021
41. Institute for Research in Fundamental Sciences, Tehran, Iran
IPM Holography Virtual Seminar
Distinguishability in Random States, Eigenstates, and Gravity
Nov. 2021
42. University of Illinois, Urbana Champaign, IL
Mathematical and Theoretical Physics Seminar
Distinguishability in Random States, Eigenstates, and Gravity
Oct. 2021
43. Harvard University, Cambridge, MA
Vishwanath Group Meeting
Distinguishability of random states and the subsystem ETH
Jul. 2021
44. Strings 2021 Gong Show, São Paulo, Brazil
Distinguishing Random States and Black Holes
Jun. 2021
45. Yukawa Institute for Theoretical Physics, Kyoto, Japan
Entanglement Meeting
Entanglement Negativity Spectrum of Random Mixed States
Feb. 2021
46. University of Illinois Urbana Champaign, Champaign IL
Condensed Matter-AMO Journal Club
Universality of Entanglement Dynamics in Integrable and Chaotic Quantum Systems
Oct. 2020
47. Massachusetts Institute of Technology, Cambridge, MA
CTP Group Meeting
 $T\bar{T}$, the entanglement wedge cross section, and the breakdown of the split property
Aug. 2020
48. Stanford University, Stanford, CA
SITP $T\bar{T}$ Group Meeting Seminar
 $T\bar{T}$, the entanglement wedge cross section, and the breakdown of the split property
Jun. 2020

49. Colgate University, Hamilton, NY Jun. 2020
Physics & Astronomy Seminar
Entanglement, Gravity, and the Holographic Principle
50. Massachusetts Institute of Technology, Cambridge, MA Apr. 2020
Condensed Matter Theory Seminar
Universality of Entanglement Dynamics
51. Albert Einstein Institute, Potsdam, Germany Dec. 2019
Gravity, Quantum Fields, & Information Seminar
Derivation of holographic negativity in $\text{AdS}_3/\text{CFT}_2$
52. AMS Special Sessions on Analysis and Geometry of Fractals, Riverside, CA Nov. 2017
Strong shape theorems in cellular automata models on fractal graphs

Teaching Experience

- Prison Teaching Initiative Instructor* 2023-Present
Princeton University, Princeton, NJ
- **PHYS-130: *Astronomy*** (Fall 2025) East Jersey State Prison
 - **PHYS-130: *Astronomy*** (Spring 2025) Northern State Prison
 - **PHYS-130: *Astronomy*** (Fall 2024) East Jersey State Prison
 - **MATH-110: *Statistics*** (Spring 2024) South Woods State Prison
 - **MATH-015: *Basic Mathematics*** (Fall 2023) South Woods State Prison
- Prison Teaching Initiative Course Coordinator* 2024-2025
Princeton University, Princeton, NJ
- **PHYS-130: *Astronomy*** (Spring 2025)
 - **PHYS-130: *Astronomy*** (Fall 2024)
- Guest Lecturer* 2024
Princeton University Department of Physics, Princeton, NJ
- **PHY 101: *Introductory Physics I***. Lecture on “Vibrational Motion and Mechanical Waves.”
- Tyler School of Art and Architecture, Philadelphia, PA
- **ART 3505: *Color***. Lecture on “The Physics of Light and Color.”
- Princeton University Teaching Transcript Program* 2022-2024
The McGraw Center for Teaching and Learning, Princeton University, Princeton, NJ
- Includes five pedagogy workshops, guest lecturing with classroom observation and analysis, and the development of an original syllabus. Completed Fall 2024.
- Reading Course Instructor* 2019
University of Chicago, Chicago, IL
- Designed and taught reading course on “*Quantum Information and Quantum Chaos*” to undergraduate physics majors at University of Chicago.
- Splash! Chicago Teacher* 2018
University of Chicago, Chicago, IL
- Designed and taught a (5 week) course “*Introduction to Quantum Mechanics*” to local high school students.
 - Designed and taught a (1 day) course “*Introduction to Quantum Mechanics and Quantum Computing*” to local high school students.
- Graduate Teaching Assistant* 2017 - 2021
Department of Physics, University of Chicago, Chicago, IL

- **PHSC 116:** *Physics for Future Presidents: Fundamental Concepts*,
- **PHSC 117:** *Physics for Future Presidents: Energy & Sustainability*,
- **PHYS 121-123:** *General Physics I-III*
- **PHYS 133:** *Waves, Optics, & Heat*
- **PHYS 142:** *Honors Electricity & Magnetism*
- **PHYS 341:** *Graduate Quantum Mechanics I* (Grader)
- **PHYS 484:** *String Theory II* (Grader)

Undergraduate Teaching Assistant and Tutor

2015-2017

Department of Physics & Astronomy, Colgate University, Hamilton, NY

- **PHYS 232:** *Introduction to Mechanics*
- **CORE 101:** *Energy and Sustainability*
- **PHYS 111:** *Fundamental Physics I*
- **ASTR 102:** *Stars, Galaxies, and the Universe*

Mentorship

Research Advisor

Bowen Ouyang - 4th year undergraduate student at Shanghai Jiao Tong University (2025)

Hyaline Chen - 3rd-4th year undergraduate Physics student at Princeton University (2024). Resulting publication. Winner of PACM Undergraduate Certificate Prize for Best Research Project

Chang-Han Chen - 4th year undergraduate Physics student at Massachusetts Institute of Technology (2020-2021). Senior Thesis

Professional Activities

Journal Referee

Physical Review Letters, Physical Review E, Nature Physics, Journal of High Energy Physics, SciPost Physics, European Physical Journal Plus, European Physical Journal C, Journal of Mathematical Physics, Quantum

Organizer

Institute for Advanced Study/Princeton Center for Theoretical Science Workshop: "Workshop on Quantum Aspects of Black Holes and Spacetime" (Dec 2025)

Princeton Center for Theoretical Science Workshop: "Random Physics" (May 2024)

Princeton Center for Theoretical Science Lunch Seminars (Fall 2023-Spring 2024)

Princeton Center for Theoretical Science Workshop: "Quantum Information, Dynamics and Ergodicity: From Many-Body Systems to Gravity" (Feb.-Mar. 2023)

Kadanoff Center for Theoretical Physics Journal Club (University of Chicago)

Kadanoff Center Group Meeting → Theory Group Meeting (University of Chicago)